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FINALTERM EXAMINATION
Fall 2008
STA301- Statistics and Probability (Session - 1)

Marks: 70

Question No: 1 (Marks: 1) - Please choose one

In a multiplication theorem P (A∩B) equals:

- ► P (A) P (B)
- ► P (A) + P (B)
- ► P (A) * P (B|A)
- ► P(B\A)*P(B)

Question No: 2 (Marks: 1) - Please choose one

The value of χ^2can never be :

- ► Zero
- ► Less than 1
- ► Greater than 1
- ► Negative

Question No: 3 (Marks: 1) - Please choose one

In the case of the sampling distribution of $\bar{X}$,is known as the finite population correction factor (fpc) is:

- ► $\sqrt{\frac{N-1}{N-n}}$

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$\sqrt{\frac{N-n}{N-1}}$

►

$\sqrt{\frac{n-N}{N-1}}$

►

► Both A & B

Question No: 4 (Marks: 1) - Please choose one

Which dispersion is used to compare variation of two series:

► C.V.

► Q.D.

► M.D.

► S.D.

Question No: 5 (Marks: 1) - Please choose one

A random variable can be generated:

► Manually

► Mechanically

► Both (a) and (b)

► Mathematically

Question No: 6 (Marks: 1) - Please choose one

The number of parameters in a Poisson distribution is (are):

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► 1

► 0

► 2

► 3

Question No: 7 (Marks: 1) - Please choose one

P (A ∩ B) =P (A/B) P (B), then A and B are:

► Dependent

► Equally Likely

► Mutually Exclusive

► Independent

Question No: 8 (Marks: 1) - Please choose one

Which one provides the basis for hypothesis testing?

► Null hypothesis

► Alternative hypothesis

► Critical value

► Test-statistics

Question No: 9 (Marks: 1) - Please choose one

The continuity correction factor is used when:

► The sample size is at least 5

► Both *nP* and *n (1-P)* are at least 30

► A continuous distribution is used to approximate a discrete distribution

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► The standard normal distribution is applied

Question No: 10 (Marks: 1) - Please choose one

$$\frac{\ell^{-0.135}\ 0.135^{1}}{1!}$$

For the Poisson distribution P(x) = the mean value is :

► 2

► 5

► 10

► 0.135

Question No: 11 (Marks: 1) - Please choose one

In a set of 10 observations the mean is 20 and the median is 15. There are 2 values that are 6, and all other values are different. What is the value mode

► 15

► 20

► 6

► 10

► 2

Question No: 12 (Marks: 1) - Please choose one

If a curve has a longer tail to the left, it is called a:

► Symmetric curve

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► positively skewed

► Negatively skewed

► J-shaped

Question No: 13 (Marks: 1) - Please choose one

A normally distributed population has known standard deviation of 1.0. the total width of a 95% confidence interval for the population mean is:

► 1.96

► 3.92

► 1.645

► Can not be determined from the given information

Question No: 14 (Marks: 1) - Please choose one

In interval estimation we always get:

► Single value

► Two values

► Range of values

► Zero

Question No: 15 (Marks: 1) - Please choose one

The best unbiased estimator for population variance σ^2 is:

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► Sample mean

► Sample median

► Sample proportion

► Sample variance

Question No: 16 (Marks: 1) - Please choose one

A normal population distribution is needed from the following test statistic:

► chi-squared

► standard error of the mean

► variance estimation

► Kendall's rank coefficient

Question No: 17 (Marks: 1) - Please choose one

When the random variable X and Y are independent, its co-variance is:

► One

► Negative

► Zero

► Positive

Question No: 18 (Marks: 1) - Please choose one

The combined distribution of more than two random variables is:

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- ► Univariate distribution
- ► Joint distribution
- ► Marginal distribution
- ► Bivariate distribution

Question No: 19 (Marks: 1) - Please choose one

By definition $f(x_i \mid y_j)$ = _____

- ► $\frac{f\left(x_i, y_j\right)}{h\left(y_j\right)}$
- ► $\frac{f\left(x_i, y_j\right)}{h\left(x_i\right)}$
- ► $\underline{f\left(x_i, y_j\right)}$
- ► $f\left(y_j\right)$

Question No: 20 (Marks: 1) - Please choose one

When f (x, y) is bivariate probability density function of continuous r.v.'s X and Y, then

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$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy$$

is equal to:

► 1

► 0

► -1

► ∞

Question No: 21 (Marks: 1)

EVENT	AMOUNT EARNED ($) x	PROBABILITY P(x)
Rain	30	0.3
No Rain	–6	a
	Total	1

Find 'a', if above is a probability distribution.

Question No: 22 (Marks: 1)

If p = 0.4 and n = 2 what is the value of σ p?

Question No: 23 (Marks: 1)

When chi- square distribution tends to normality?

Question No: 24 (Marks: 2)

Write down the significance of experimental design.

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Question No: 25 (Marks: 2)

Write down the formulas that should be used to calculate Mean, Variance and Standard deviation of a continuous random variable:

Question No: 26 (Marks: 2)

Define normal distribution.

Question No: 27 (Marks: 3)

Describe the paired t-test.

Question No: 28 (Marks: 3)

Write down the any two properties of probability density function.

Question No: 29 (Marks: 5)

Describe the 'standard error' in sampling distribution of proportions.

Question No: 30 (Marks: 5)

Find the expected value of the random variable X having the p.d.f.

$f(x) = 2(1 - x), \quad 0 < x < 1$

$= 0$, elsewhere.

Question No: 31 (Marks: 5)

A random sample of size n is drawn from normal population with mean 5 and variance σ^2. Answer the following:

If s=15, $\bar{x}$ =14 and t=3, what is values of n?

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Question No: 32 (Marks: 10)

An electrical firm manufactures light bulbs that have a length of life that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours.

Question No: 33 (Marks: 10)

A population consists of N=4 numbers 1, 3, 4 and 5. Find proportion $\hat{p}$ of odd numbers in all possible samples of size n=3 that can be draw without replacement from the population. Construct the sampling distribution of odd numbers and find mean and variance of this distribution.

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